Fatine Kassabi

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Data scientist with an M.Sc. in Data Science (Lille, 2026) and two machine learning internships. Builds models end to end, from Bayesian and causal modeling to LLM-based retrieval, shipped as tested Python code with APIs, dashboards, and CI. Seeking a Data Scientist or ML Engineer role in industry.

Skills

Statistics: Bayesian inference, MCMC, causal inference, uplift modeling, A/B testing
Programming & ML: Python (pandas, NumPy, scikit-learn, LightGBM, PyTorch), SQL, R
LLMs & NLP: Hugging Face Transformers, Qwen2.5, Llama 3.3, RAG, LangChain, ChromaDB, prompt engineering
Data & Cloud: Snowflake, AWS (Lambda, Step Functions, SNS, S3)
Engineering & MLOps: FastAPI, Streamlit, Docker, Git, GitHub Actions (CI), pytest, Linux, Bash
Languages: French (C1), English (C1), Arabic (native)

Professional Experience

AI Intern (final-year M.Sc. internship): LLM-based retrieval and Bayesian inference Mar. – Sept. 2026
Laboratoire Paul Painlevé (University of Lille, CNRS) *Villeneuve-d'Ascq, France*

- Built a Bayesian retrieval system that infers which of 31,114 Chinese idioms best summarizes a given text, combining a Qwen2.5 LLM likelihood with an idiom-frequency prior; served it through a FastAPI endpoint.
- Designed an embedding-guided Metropolis–Hastings sampler to approximate the posterior: on held-out texts, its top pick matched the exact posterior's top idiom 22.8% of the time, vs. 17.2% for a uniform-proposal sampler (chance: 0.003%).
- Tested three distance measures across all model layers to find a signal for idiom–text fit; the best (Euclidean distance, layer 23) picked the correct idiom over a wrong one 72% of the time (chance: 50%).

Machine Learning Intern: deep learning for climate downscaling Mar. – Sept. 2025
International Water Research Institute (IWRI), Mohammed VI Polytechnic University (UM6P) *Benguerir, Morocco*

- Developed a precipitation downscaling pipeline (preprocessing, feature engineering) on 27 years of climate data (LMDZ-IPSL model outputs, ERA5 reanalysis), from 250 km to 35 km resolution.
- Benchmarked a Vision Transformer and a convolutional autoencoder (PyTorch, GPU) against Random Forest and SVM baselines (scikit-learn), evaluating pointwise accuracy (RMSE, R^2) and spatial bias across grid cells.
- Found that Random Forest ($R^2 = 0.88$), the best baseline model, outperformed the Vision Transformer ($R^2 = 0.85$), the best deep-learning model.

Academic & Personal Projects

Uplift modeling for marketing optimization (Criteo, 3 million users) June – Aug. 2026
Personal project *github.com/FatineKa/retail-media-uplift | live dashboard*

- Built and compared four uplift models combining T-learner, S-learner, and class-transformation approaches with LightGBM and logistic regression; selected the best model on a validation set and evaluated it with **Qini curves** on 900,000 held-out users.
- Assessed the profit gain over propensity targeting with a **paired bootstrap**: at a 2% budget, uplift targeting generated an additional $3,034, although the difference was not statistically significant.
- Deployed an automated data quality gate (Snowflake, AWS Lambda) and an interactive Streamlit dashboard.

LLM/RAG system: personalized academic guidance Jan. – Mar. 2026
Academic project, University of Lille *github.com/FatineKa/orientation-rag*

- Engineered a two-stage RAG pipeline over 3,354 degree programs: ChromaDB vector search with multilingual embeddings and metadata filters extracted from queries and PDF forms, then generation constrained to the retrieved programs.
- Integrated Llama 3.3, GPT-4o-mini, and Mistral via LangChain behind a Streamlit interface.

Education

M.Sc. Statistical & Numerical Engineering (Data Science) 2025 – 2026
University of Lille, Faculty of Science and Technology *Villeneuve-d'Ascq, France*

M.Sc. Mathematical Modeling & Data Science 2023 – 2025
Abdelmalek Essaâdi University, Faculty of Sciences and Techniques *Tangier, Morocco*

B.Sc. Statistical Engineering 2020 – 2023
Abdelmalek Essaâdi University, Faculty of Sciences and Techniques *Tangier, Morocco*